

Product ORYX MUSK

Print Date 19.12.2018

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1. Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Trade name : ORYX MUSK
IFF Code : 5663455
SDS Number : 300000734098

1.2 Relevant identified uses of the substance or mixture and uses advised against

Use of the Substance/Mixture : Fragrance for consumer product

1.3 Details of the supplier of the safety data sheet

Company : IFF (NEDERLAND) BV
ZEVENHEUVELENWEG 60
5048 AN TILBURG
Netherlands
Telephone : +31134642211
Telefax : +31134636032
E-mail address : sds@iff.com
Responsible/issuing person

1.4 Emergency telephone number

Refer to section 16 for country specific emergency contact number.

2. Hazards identification

2.1 Classification of the substance or mixture

Classification (REGULATION (EC) No 1272/2008)

Skin irritation, Category 2	H315: Causes skin irritation.
Eye irritation, Category 2	H319: Causes serious eye irritation.
Skin sensitisation, Category 1	H317: May cause an allergic skin reaction.
Carcinogenicity, Category 2	H351: Suspected of causing cancer.
Reproductive toxicity, Category 2	H361: Suspected of damaging fertility or the unborn child.
Chronic aquatic toxicity, Category 1	H410: Very toxic to aquatic life with long lasting effects.

2.2 Label elements

Labelling (REGULATION (EC) No 1272/2008)

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Hazard pictograms



Signal word

: Warning

Hazard statements

:	H315 H317 H319 H351 H361 H410	Causes skin irritation. May cause an allergic skin reaction. Causes serious eye irritation. Suspected of causing cancer. Suspected of damaging fertility or the unborn child. Very toxic to aquatic life with long lasting effects.
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Precautionary statements

:	Prevention: P201 P261 P273 P280 Response: P308 + P313 P362 + P364	Obtain special instructions before use. Avoid breathing dust/ fume/ gas/ mist/ vapours/ spray. Avoid release to the environment. Wear protective gloves/ protective clothing/ eye protection/ face protection. IF exposed or concerned: Get medical advice/ attention. Take off contaminated clothing and wash it before reuse.
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Hazardous components which must be listed on the label:

- 68155-66-8, 54464-57-2, 68155-67-9, 54464-59-4 Tetramethyl Acetyloctahydronaphthalenes
- 1205-17-0 α -methyl-1,3-benzodioxole-5-propionaldehyde
- 81-14-1 musk ketone
- 33704-61-9 Dihydro pentamethylindanone
- 23696-85-7 1-(2,6,6-trimethyl-1,3-cyclohexadien-1-yl)-2-buten-1-one

2.3 Other hazards

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None reasonably foreseeable.

3. Composition/information on ingredients

3.1 Substances

Not applicable, product is a mixture.

3.2 Mixtures

Hazardous components

Chemical name	CAS-No. EC-No. Registration number	Classification (REGULATION (EC) No 1272/2008)	Concentration [%]
1,3,4,6,7,8-hexahydro-4,6,6,7,8,8-hexamethylindeno[5,6-c]pyran	1222-05-5 214-946-9 01-2119488227-29	Aquatic Chronic1; H410 Aquatic Acute1; H400	10 - 20
Tetramethyl Acetyloctahydronaphthalenes	68155-66-8, 54464-57-2, 68155-67-9, 54464-59-4 915-730-3 01-2119489989-04	Skin Irrit.2; H315 Skin Sens.1B; H317 Aquatic Chronic1; H410	10 - 20
tetrahydro-2-isobutyl-4-methylpyran-4-ol, mixed isomers (cis and trans)	63500-71-0 603-101-00-3 01-0000015458-64	Eye Irrit.2; H319	10 - 20
$\alpha,\beta,2,2,3$ -pentamethylcyclopent-3-ene-1-butanol	65113-99-7 265-453-0	Aquatic Chronic2; H411 Eye Irrit.2; H319	3 - 10
α -methyl-1,3-benzodioxole-5-propionaldehyde	1205-17-0 214-881-6	Skin Sens.1B; H317 Repr.2; H361 STOT RE2; H373 Aquatic Chronic2; H411	3 - 10
musk ketone	81-14-1 201-328-9	Carc.2; H351 Aquatic Chronic1; H410 Aquatic Acute1; H400	2,5 - 10
1-(5,6,7,8-tetrahydro-3,5,5,6,8,8-hexamethyl-2-naphthyl)ethan-1-one	21145-77-7 244-240-6	Acute Tox.4; H302 Aquatic Chronic1; H410 Aquatic Acute1; H400	1 - 2,5
2-ethyl-4-(2,2,3-trimethyl-3-cyclopenten-1-yl)-2-buten-1-	28219-61-6 701-122-3	Eye Irrit.2; H319 Aquatic Chronic2; H411	1 - 2,5

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ol	01-2119529224-45		
2-phenylethanol	60-12-8 200-456-2	Eye Irrit.2; H319 Acute Tox.4; H302	1 - 3
1,4-dioxacyclohexadecane-5,16-dione	54982-83-1 259-423-6 01-2119524000-64	Aquatic Acute1; H400 Aquatic Chronic3; H412	0,25 - 1
Dihydro pentamethylindanone	33704-61-9 251-649-3 01-2119977131-40	Aquatic Chronic2; H411 Skin Sens.1B; H317 Eye Irrit.2; H319 Skin Irrit.2; H315	0,25 - 1
[3R-(3.alpha.,3a.beta.,7.beta.,8a.alpha.)]-2,3,4,7,8,8a-hexahydro-3,6,8,8-tetramethyl-1H-3a,7-methanoazulene	469-61-4 207-418-4	Asp. Tox.1; H304 Aquatic Chronic1; H410 Aquatic Acute1; H400	0,25 - 1
tetrahydro-4-methyl-2-(2-methylprop-1-enyl)pyran	16409-43-1 240-457-5 01-2119976300-42	Skin Irrit.2; H315 Eye Irrit.2; H319 Repr.2; H361	0,1 - 0,3
1-(2,6,6-trimethyl-1,3-cyclohexadien-1-yl)-2-buten-1-one	23696-85-7 245-833-2	Skin Sens.1A; H317 Aquatic Chronic2; H411 Skin Irrit.2; H315	0,025 - 0,1

For the full text of the R-phrases mentioned in this Section, see Section 16.

Components with workplace control parameters

Components	EC-No.	Concentration [%]
2-phenylethanol	200-456-2	0 - 10

4. First aid measures

4.1 Description of first aid measures

- General advice : Take Hazard and Precautionary phrases (section 2) into account.
- If inhaled : Remove from exposure site to fresh air and keep at rest. If victim is unconscious, remove foreign bodies from the mouth. If victim has stopped breathing, give artificial respiration. Obtain medical advice.

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- | | |
|-------------------------|--|
| In case of skin contact | : Remove contaminated clothes. Wash thoroughly with water (and soap). Contact physician if symptoms persist. |
| In case of eye contact | : Flush immediately with water for at least 15 minutes. Contact physician if symptoms persist. |
| If swallowed | : Rinse mouth with water and obtain medical advice. |

4.2 Most important symptoms and effects, both acute and delayed

- | | |
|----------|-----------------------------|
| Symptoms | : No information available. |
| Risks | : No information available. |

4.3 Indication of any immediate medical attention and special treatment needed

- | | |
|-----------|-----------------------------|
| Treatment | : No information available. |
|-----------|-----------------------------|

5. Firefighting measures

5.1 Extinguishing media

- | | |
|--------------------------------|---|
| Suitable extinguishing media | : Carbondioxide, dry chemical, foam. |
| Unsuitable extinguishing media | : Do not use a direct waterjet on burning material. |

5.2 Special hazards arising from the substance or mixture

- | | |
|--------------------------------------|-----------------------------|
| Specific hazards during firefighting | : Water may be ineffective. |
|--------------------------------------|-----------------------------|

5.3 Advice for firefighters

- | | |
|---------------------|--|
| Further information | : Standard procedure for chemical fires. |
|---------------------|--|

6. Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

- | | |
|----------------------|--|
| Personal precautions | : Avoid inhalation and contact with skin and eyes. A self-contained breathing apparatus is recommended in case of a major spill. |
|----------------------|--|

6.2 Environmental precautions

- | | |
|---------------------------|---|
| Environmental precautions | : Keep away from drains, surface- and groundwater and soil. |
|---------------------------|---|

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6.3 Methods and materials for containment and cleaning up

- Methods for cleaning up : Clean up spillage promptly. Remove ignition sources. Provide adequate ventilation. Avoid excessive inhalation of vapours. Gross spillages should be contained by use of sand or inert powder and disposed of according to the local regulations.

6.4 Reference to other sections

Prevent spreading over a wide area (e.g. by containment or oil barriers).

7. Handling and storage

7.1 Precautions for safe handling

- Advice on safe handling : Avoid excessive inhalation of concentrated vapors. Follow good manufacturing practices for housekeeping and personal hygiene. Wash any exposed skin immediately after any chemical contact, before breaks and meals, and at the end of each work period. Contaminated clothing and shoes should be thoroughly cleaned before re-use.
- If appropriate, procedures used during the handling of this material should also be used when cleaning equipment or removing residual chemicals from tanks or other containers, especially when steam or hot water is used, as this may increase vapor concentrations in the workplace air. Where chemicals are openly handled, access should be restricted to properly trained employees.
- Keep all heated processes at the lowest necessary temperature in order to minimize emissions of volatile chemicals into the air.

- Advice on protection against fire and explosion : Keep away from ignition sources and naked flame.

7.2 Conditions for safe storage, including any incompatibilities

- Requirements for storage areas and containers : Store in a cool, dry, ventilated area away from heat sources. Keep containers upright and tightly closed when not in use.

7.3 Specific end use(s)

- Specific use(s) : No information available.

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8. Exposure controls/personal protection

8.1 Control parameters

Components with workplace control parameters

Components	EC-No.	Exposure limit(s)
2-phenylethanol	200-456-2	TWA

DNEL

1,3,4,6,7,8-hexahydro-
4,6,6,7,8,8-
hexamethylindeno[5,6-c]pyran

: End Use: Workers
Exposure routes: Skin contact
Potential health effects: Long-term systemic effects
Exposure time: 8 h
Value: 60 mg/kg bw/day

End Use: Workers
Exposure routes: Inhalation
Potential health effects: Long-term systemic effects
Exposure time: 8 h
Value: 22 mg/m3

End Use: General population
Exposure routes: Inhalation
Potential health effects: Long-term systemic effects
Exposure time: 8 h
Value: 6,5 mg/m3

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End Use: General population
Exposure routes: Skin contact
Potential health effects: Long-term systemic effects
Exposure time: 8 h
Value: 36 mg/kg bw/day

End Use: General population
Exposure routes: Ingestion
Potential health effects: Long-term systemic effects
Exposure time: 8 h
Value: 3,8 mg/kg bw/day

1-(1,2,3,4,5,6,7,8-Octahydro-
2,3,8,8-tetramethyl-2-
naphthyl)ethan-1-one

: End Use: Workers
Exposure routes: Skin contact
Potential health effects: Long-term local effects
Exposure time: 8 h
Value: 0,1011 mg/cm2

End Use: Workers
Exposure routes: Skin contact
Potential health effects: Long-term systemic effects
Exposure time: 8 h
Value: 1,73 mg/kg bw/day

End Use: Workers
Exposure routes: Inhalation
Potential health effects: Long-term systemic effects
Exposure time: 8 h
Value: 1,76 mg/m3

1,4-dioxacyclohexadecane-5,16- : End Use: Workers

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dione

Exposure routes: Skin contact
Potential health effects: Long-term systemic effects
Exposure time: 8 h
Value: 25,8 mg/kg bw/day

End Use: Workers
Exposure routes: Inhalation
Potential health effects: Long-term systemic effects
Exposure time: 8 h
Value: 182 mg/m3

End Use: General population
Exposure routes: Skin contact
Potential health effects: Long-term systemic effects
Exposure time: 8 h
Value: 15,5 mg/kg bw/day

End Use: General population
Exposure routes: Inhalation
Potential health effects: Long-term systemic effects
Exposure time: 8 h
Value: 15,9 mg/m3

End Use: General population
Exposure routes: Ingestion
Potential health effects: Long-term systemic effects
Exposure time: 8 h
Value: 15,5 mg/kg bw/day

PNEC
1,3,4,6,7,8-hexahydro-

: Fresh water

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4,6,6,7,8,8-hexamethylindeno[5,6-c]pyran

Value: 0,0044 mg/l

Marine water

Value: 0,00044 mg/l

Fresh water sediment

Value: 2 mg/kg dry weight (d.w.)

Marine sediment

Value: 0,394 mg/kg dry weight (d.w.)

Soil

Value: 0,31 mg/kg dry weight (d.w.)

1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one

: Fresh water

Value: 0,0028 mg/l

Marine water

Value: 0,00028 mg/l

Fresh water sediment

Value: 3,73 mg/kg dry weight (d.w.)

Marine sediment

Value: 0,75 mg/kg dry weight (d.w.)

Soil

Value: 0,705 mg/kg dry weight (d.w.)

1,4-dioxacyclohexadecane-5,16-dione

: Fresh water

Value: 0,00088 mg/l

Marine water

Value: 0,000088 mg/l

Fresh water sediment

Value: 0,162 mg/kg dry weight (d.w.)

Marine sediment

Value: 0,0162 mg/kg dry weight (d.w.)

Soil

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Value: 0,0318 mg/kg dry weight (d.w.)

Sewage treatment plant

Value: 1,8 mg/l

8.2 Exposure controls

Engineering measures

Where appropriate, use closed systems to transfer and process this material.

If appropriate, isolate mixing rooms and other areas where this material is used or openly handled. Maintain these areas under negative air pressure relative to the rest of the plant.

Personal protective equipment

Respiratory protection : Use local exhaust ventilation around open tanks and other open sources of potential exposures in order to avoid excessive inhalation, including places where this material is openly weighed or measured. In addition, use general dilution ventilation of the work area to eliminate or reduce possible worker exposures. No respiratory protection is required during normal operations in a workplace where engineering controls such as adequate ventilation, etc. are sufficient.

If engineering controls and safe work practices are not sufficient, an approved, properly fitted respirator with organic vapor cartridges or canisters and particulate filters should be used:

- a) while engineering controls and appropriate safe work practices and/or procedures are being implemented; or
- b) during short term maintenance procedures when engineering controls are not in normal operation or are not sufficient; or
- c) if normal operational workplace vapor concentration in the air is increased due to heat ;
- d) during emergencies; or
- e) if engineering controls and operational practices are not sufficient to reduce airborne concentrations below an established occupational exposure limit.

Hand protection : Avoid skin contact. Use chemically resistant gloves.

Eye protection : Use tight-fitting goggles, face shield or safety glasses with side shields if eye contact might occur.

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Hygiene measures : To the extent deemed appropriate, implement pre-placement and regularly scheduled ascertainment of symptoms and spirometry testing of lung function for workers who are regularly exposed to this material.
To the extent deemed appropriate, use an experienced air sampling expert to identify and measure volatile chemicals that could be present in the workplace air to determine potential exposures and to ensure the continuing effectiveness of engineering controls and operational practices to minimize exposure.

Environmental exposure controls

General advice : Keep away from drains, surface- and groundwater and soil.

9. Physical and chemical properties

9.1 Information on basic physical and chemical properties

Appearance	: liquid
Odour	: conforms to standard
Odour Threshold	: not determined
Flash point	: 123 °C
Lower explosion limit	: not determined
Upper explosion limit	: not determined
Flammability (solid, gas)	: not determined
Oxidizing properties	: not determined
Auto-ignition temperature	: not determined
pH	: not determined
Melting point	: not determined
Boiling point	: not determined
Vapour pressure	: < 0,02 hPa Calculated
Density	: not determined
Water solubility	: not determined
Partition coefficient: n-octanol/water	: not determined
Solubility in other solvents	: not determined
Viscosity, dynamic	: not determined
Viscosity, kinematic	: not determined
Relative vapour density	: not determined
Evaporation rate	: not determined

9.2 Other information

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Refractive index : not determined
Relative density : 1,0062 - 1,0162

10. Stability and reactivity

10.1 Reactivity

No hazards to be specially mentioned.

10.2 Chemical stability

Stable under normal conditions.

10.3 Possibility of hazardous reactions

Hazardous reactions : Note: Presents no significant reactivity hazard, by itself or in contact with water. Avoid contact with strong acids, alkali or oxidizing agents.

10.4 Conditions to avoid

Conditions to avoid : Direct sources of heat.

10.5 Incompatible materials

Materials to avoid : Avoid contact with strong acids, alkali or oxidizing agents.

10.6 Hazardous decomposition products

Hazardous decomposition products : Carbon monoxide and unidentified organic compounds may be formed during combustion.

11. Toxicological information

11.1 Information on toxicological effects

Acute toxicity

Acute oral toxicity : Acute toxicity estimate: > 2.000 mg/kg
Method: Calculation method

Skin corrosion/irritation

1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one : Species: human
Result: Skin irritation
Method: OECD 439

Serious eye damage/eye irritation

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tetrahydro-2-isobutyl-4-methylpyran-4-ol, mixed isomers (cis and trans) : Species: Rabbit
Result: Eye irritation

Respiratory or skin sensitisation

1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one : LLNA
Species: Mouse
Result: Causes sensitisation.
Method: OECD 429

Germ cell mutagenicity

Information on ingredients with chronic toxicity hazard(s) is available upon request.

Carcinogenicity

Information on ingredients with chronic toxicity hazard(s) is available upon request.

Reproductive toxicity

Information on ingredients with chronic toxicity hazard(s) is available upon request.

Target Organ Systemic Toxicant - Single exposure

Information on ingredients with chronic toxicity hazard(s) is available upon request.

Target Organ Systemic Toxicant - Repeated exposure

Information on ingredients with chronic toxicity hazard(s) is available upon request.

Aspiration hazard

No test data available. Ingredients with an aspiration hazard are identified in section 3.

12. Ecological information

12.1 Toxicity

Toxicity to fish
1,3,4,6,7,8-hexahydro-4,6,6,7,8,8-hexamethylindeno[5,6-c]pyran : LC50: 0,452 mg/l
Exposure time: 21 d
Species: Lepomis macrochirus (Bluegill sunfish)
Method: OECD Test Guideline 204

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Toxicity to daphnia and other aquatic invertebrates

1,3,4,6,7,8-hexahydro- : EC50: 0,9 mg/l
4,6,6,7,8,8- Exposure time: 48 h
hexamethylindeno[5,6-c]pyran Species: Daphnia magna (Water flea)
Method: OECD Test Guideline 202

Toxicity to algae

1,3,4,6,7,8-hexahydro- : ErC50: > 0,854 mg/l
4,6,6,7,8,8- Exposure time: 72 h
hexamethylindeno[5,6-c]pyran Species: Pseudokirchneriella subcapitata (green algae)
Method: OECD Test Guideline 201

EbC50: 0,723 mg/l
Exposure time: 72 h
Species: Pseudokirchneriella subcapitata (green algae)
Method: OECD Test Guideline 201

Toxicity to fish (Chronic toxicity)

1,3,4,6,7,8-hexahydro- : NOEC: 0,068 mg/l
4,6,6,7,8,8- Exposure time: 36 d
hexamethylindeno[5,6-c]pyran Species: Pimephales promelas (fathead minnow)
Method: OECD 210

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)

1,3,4,6,7,8-hexahydro- : NOEC: 0,111 mg/l
4,6,6,7,8,8- Exposure time: 21 d
hexamethylindeno[5,6-c]pyran Species: Daphnia magna (Water flea)
Method: OECD 211

12.2 Persistence and degradability

Biodegradability

1,3,4,6,7,8-hexahydro- : Result: Not readily biodegradable.
4,6,6,7,8,8- 2 %
hexamethylindeno[5,6-c]pyran Method: Modified Sturm Test

12.3 Bioaccumulative potential

Bioaccumulation

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1,3,4,6,7,8-hexahydro- : Species: Lepomis macrochirus (Bluegill sunfish)
4,6,6,7,8,8- Exposure time: 28 d
hexamethylindeno[5,6-c]pyran Bioconcentration factor (BCF): 1.584
Method: OECD 305

12.4 Mobility in soil

Mobility : No data available

Distribution among environmental compartments

1,3,4,6,7,8-hexahydro- : Log Koc: 4,39
4,6,6,7,8,8-
hexamethylindeno[5,6-c]pyran

12.5 Results of PBT and vPvB assessment

This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

12.6 Other adverse effects

Additional ecological : There is no data available for this product.
information

13. Disposal considerations

13.1 Waste treatment methods

Product : Dispose of according to local regulations. Avoid disposing into drainage systems and into the environment.

Contaminated packaging : Empty containers should be taken to an approved waste handling site for recycling or disposal.

14. Transport information

ADR

UN number : 3082
Description of the goods : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
(HEXAMETHYLINDANOPYRAN)
Labels : 9
Packing group : III
Environmentally hazardous : yes

IATA

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UN number	: 3082
Description of the goods	: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (HEXAMETHYLINDANOPYRAN)
Labels	: 9
Packing group	: III
Environmentally hazardous	: yes
IMDG	
UN number	: 3082
Description of the goods	: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (HEXAMETHYLINDANOPYRAN)
Labels	: 9
Packing group	: III
Marine pollutant	: yes (HEXAMETHYLINDANOPYRAN)
Special precautions for user	: No special precautions required.

15. Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

Water contaminating class (Germany)	: WGK 3highly water endangering

15.2 Chemical safety assessment

A Chemical Safety Assessment is not required for this substance.

16. Other information

Full text of H-Statements referred to under sections 2 and 3.

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H302	Harmful if swallowed.
H304	May be fatal if swallowed and enters airways.
H315	Causes skin irritation.
H317	May cause an allergic skin reaction.
H319	Causes serious eye irritation.
H351	Suspected of causing cancer.
H361	Suspected of damaging fertility or the unborn child.
H373	May cause damage to organs through prolonged or repeated exposure.
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.
H411	Toxic to aquatic life with long lasting effects.
H412	Harmful to aquatic life with long lasting effects.

Further information

In December 2003, the National Institute for Occupational Safety and Health ("NIOSH") published an Alert on preventing lung disease in workers who use or make flavorings [NIOSH Publication Number 2004-110].

In August 2004, the United States Flavor and Extract Manufacturers Association (FEMA) issued a report entitled "Respiratory Safety in the Flavor Manufacturing Workplace".

Both of these reports provide recommendations for reducing employee exposure and for medical surveillance in the workplace. The recommendations in these reports are generally applicable to the use of any chemical in the workplace and you are strongly urged to review both of these reports.

The report published by FEMA also contains a list of "high priority" chemicals. If any of these chemicals are present in this product at a concentration $\geq 1.0\%$ due to an intentional addition by IFF, the chemical(s) will be identified in this safety data sheet.

According to Regulation (EC) No. 1907/2006 the information in this safety data sheet is based on the properties of the material known to IFF at the time the data sheet was issued. The safety data sheet is intended to provide information for a health and safety assessment of the material and the circumstances, under which it is packaged, stored or applied in the workplace. For such a safety assessment International Flavors & Fragrances holds no responsibility. This document is not intended for quality assurance purposes.

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Czech Republic	+420 224 919 293 / +420 224 915 402
Denmark	+45 82 12 12 12
Estonia	16662 (National), International (+372) 626 93 90

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